

Box 27-3 Stroking

Stroking is often used initially in a massage and on most areas. Some consider this a form of effleurage.

Uses

Relax the dog.
Introduce touch.
Sedate (slow stroking) the dog.
Stimulate (brisk stroking) the dog.
Decrease muscle tone (slow).

When to Use

At the beginning or end of massage
When only limited time is available, on a daily basis
To calm and relax in a stressful situation
Any time

Application

Stroking is usually applied from the proximal limb toward the distal limb.
Begin at the top of the area to be worked.
Place the entire hand in contact with the skin.
Maintain a gentle but firm pressure.



Figure 27-28 Stroking.



Figure 27-29 Effleurage.

Box 27-4 Effleurage

Effleurage is often performed initially in a massage on most areas of the body, particularly on the limbs and when swelling is present. It is the most common type of massage used, and often it is used in an evaluative manner to assess for tissue tightness, presence of muscle spasm, and so forth.

Uses

Increase venous and lymphatic return.
Aid removal of chemical irritants.
Improve mobility between tissues.
Stretch muscle fibers.
Decrease muscle tone.

Note: If used deeply, it may be used to increase muscle tone.

When to Use

At the start and end of a session, and often in between various techniques
Should be performed toward the lymph nodes
May be performed using one or both hands, fingers or thumbs

Application

Begin at the distal part to be effleuraged.
Make contact with the skin and apply even pressure to sink into the superficial tissues.
Make a sweeping movement to the top of the area, molding to the contours and maintaining the same depth of pressure throughout the stroke.
Finish the stroke over the lymph node, then remove the hands and reposition at the start of the next stroke. If the stroke is not to the lymph nodes, continue to the nearest body part.
Bring the hands toward you as you work, using the heel of your hand.
Watch that the fingers and palm of the hand do not lose contact with the patient.
Maintain an even depth of pressure while the hands mold to the body.
Overlapping strokes are used, continuing until the entire body region is covered.
Note: In a small area, use only one hand, finger or thumb.

very effective (Box 27-5). The palm or heel of the hand is placed over the muscle and pressure is applied. This pressure is maintained for 15 seconds before moving on to another area. This maneuver should be repeated several times. By restricting blood flow and then releasing pressure, the circulation increases, and this is thought to help to decrease tone (Figure 27-30).

Holding and Placing

An area of increased tone or spasm may be relaxed by placing a hand over the area. This technique traps heat in the area, increasing the local circulation and promoting local relaxation (Figure 27-31).

Percussion (Clapping, Hacking, Pounding)

If there is a general area of increased tone, such as in the caudal thigh or gluteal muscles, clapping or hacking

Box 27-5 Compression and Wringing

Compressions can be used all over and are very effective on large muscles. Wringing is often very relaxing and has a great sedative effect.

Uses

Aid venous and lymphatic return.
 Help with removal of chemical irritants.
 Increase the mobility and length of fibrous tissue.
 Restore mobility between tissue surfaces.
 Aid tissue fluid mobility.
 Increase extensibility and strength of connective tissue.
 May also trigger skin reflexes and others linked to it.

Features

These techniques compress the soft tissues, then they squeeze or roll them (wringing), working to the tissue end.

When to Use

Use on superficial tissue, ligaments, or muscles to soften up large areas.

Application of Compressions

Use one hand or both, or pads, fingers or thumbs.
 Begin at the proximal end of the tissue and move distally.
 Contact the skin and compress the tissue.
 Skin is moved on the underlying tissue; there is no glide.
 The hands or digits or back of fist are used in a circular motion; this will initiate a slight stretch behind, and skin wrinkling ahead.
 If using right and left hand, the right moves clockwise and the left counterclockwise.
 Pressure is applied to deeper tissues when tissues are compressed. It is a similar movement to that used to open a childproof medicine bottle: press down initially, twisting as you do so, then release.
 To go from one completed compression to the next, slide the hands across.

Application of Wringing

Wringing is applied using two hands.

- One brings the tissue toward you, the other moves it away. Hold for 2-5 seconds, then slowly release.
- The hand then moves to take up position on the adjacent tissue.

techniques may be employed (Box 27-6). This technique may also be used in areas of muscle weakness.

Clapping is performed by using a cupped hand. Gently but firmly, a clap is applied to the area (Figure 27-32). As the hands fall onto the muscle mass, a hollow sound should be heard. Hacking is performed by alternately using open hands, with the ulnar border of the hands falling vertically on the muscle mass (Figure 27-33). In both cases the techniques are performed with light pressure initially, progressing to medium pressure. Both clapping and hacking improve general circulation in the areas treated, promoting relaxation.



Figure 27-30 Compressions applied to tissues.



Figure 27-31 Placing technique over tissue.



Figure 27-32 Clapping of tissue.

Trigger Point Therapy

Often small areas of spasm may be felt within a muscle belly. Such nodules may be treated using trigger point therapy. The nodules are located and ischemic compression is applied using one or two fingers. The compression is held for approximately 20 seconds and released for 10

Box 27-6 Percussion

Percussion consists of several techniques, the three most important being hacking, pounding, and clapping.

Uses

- Stimulate local circulation.
- Provoke muscle and tendon reflexes.
- Provoke a general stimulatory effect.
- Stimulate muscle tone.
- Assistance in clearing airways (coupage).

Features

Light percussion has an effect on the superficial tissues; heavier application affects deeper layers and must not be used over organs.

Application of Clapping (see Figure 27-32)

Hold the fingers and thumbs closed together to form a relaxed, cup shape. The elbows should be flexed and the arms abducted. The arms should be alternately flexed and extended so that the borders of the hands and fingers stroke the skin. The strokes are rapid, light, and brisk. Air is trapped between the hands and the skin, producing a hollow sound as contact is made. The entire treatment area should be covered.

Application of Hacking (see Figure 27-33)

Hold the arms away from your sides, and flex the elbows approximately 90 degrees, with the wrists fully stretched backward (extended) and fingers relaxed. The shoulders, where possible, should be over the area.

The border of the little finger and hand strike the skin alternately, lightly and rapidly. The movement comes from the wrist, and it involves a full side-to-side movement of the wrist in this position.

When doing very light hacking, only the fingers need strike the skin.

Application of Pounding

The arms should be held out, with elbows flexed to near 90 degrees. The wrists are extended with fingers flexed loosely into fists. The hips should be flexed to enable the shoulders to be over the area being treated. The medial borders of the fifth fingers strike the skin alternately and rapidly. The movement is at the radioulnar joint, which pronates and supinates.

seconds before compression is reapplied. Generally three or four repetitions may be required (Figure 27-34).

Deep Transverse Friction

Scar tissue forms within muscles as a result of injury. Deep transverse frictions may be applied to help reduce the scarring (Box 27-7). These techniques were developed by Cyriax.²⁶ Using the index and middle fingers at a 90-degree angle to the muscle fibers, pressure is applied perpendicular to the direction of the fibers. This is repeated 10 times and performed in sets of 3 to 10. It is a good practice to



Figure 27-33 Hacking of tissue.



Figure 27-34 Compression of trigger point.

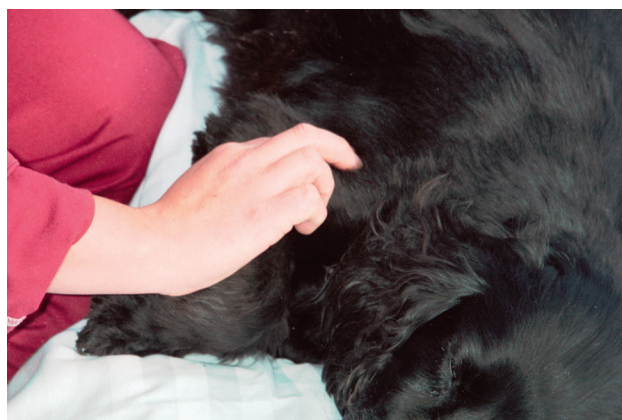


Figure 27-35 Deep transverse friction massage.

complete this technique with some gentle massage and stretching to relax the dog, as deep transverse friction may be uncomfortable (Figure 27-35).

Passive Movements

To conclude the treatment session, when all the muscles are warm and relaxed, passive movements similar to those done in the weight-bearing position to evaluate the patient

Box 27-7 Friction Massage**Traditional Friction Massage***Uses*

Stimulate local circulation.
 Aid in removal of chemical irritants.
 Restore mobility between tissue interfaces.
 Restore mobility to specific anatomic structures.

Features

Performed with the tip of the thumb or fingers making small rotary movements, the pressure being appropriate to the desired tissue interface. Lubricants are not used as there is no glide on the skin.

Application of the Friction

The tip of the thumb or fingers should be placed on the skin, over the structure to be treated. The tissues are compressed to the depth of the structures being treated. Small rotary movements are performed on the structures while maintaining constant pressure, but there is no glide on the skin. The superficial tissues are moved on the underlying structures.

Deep Transverse Friction*Uses*

Stimulate local circulation.
 Promote reactive hyperemia.
 Restore mobility between tissue interfaces.
 Restore mobility to specific anatomic structures.

Features

Performed with the tip of the thumb or fingers making small rotary movements, the pressure being appropriate to the desired tissue interface. Lubricants are not used as there is no glide on the skin.

Application of Friction (see Figure 27-35)

The tip of the thumb or fingers should be placed on the skin, over the structure to be treated. The tissues are compressed to the depth of the structures being treated. Small rotary movements are performed on the structures while maintaining constant pressure, but there is no glide on the skin. The superficial tissues are moved on the underlying structures.

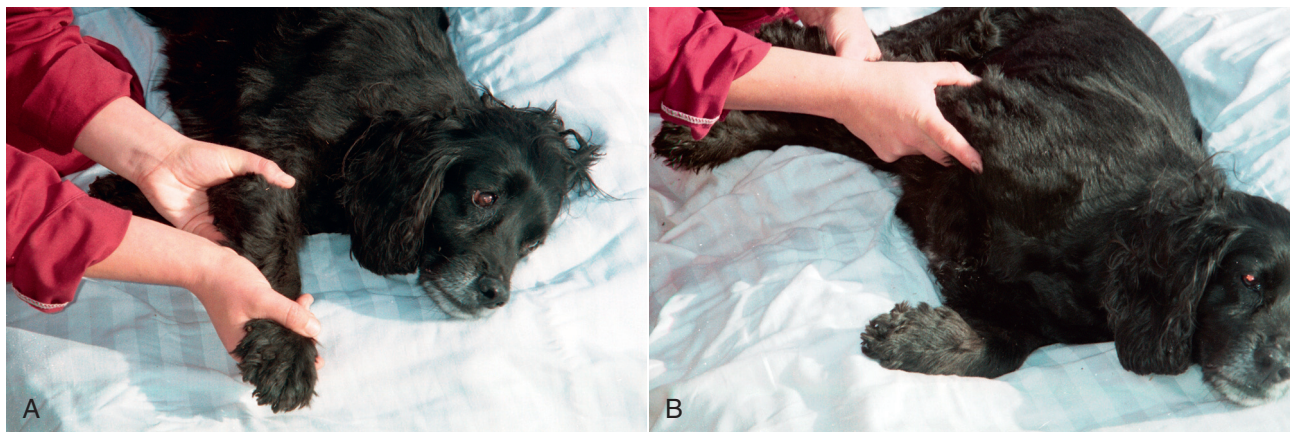


Figure 27-36 A, Forelimb mobilization with limb fully protracted. B, Forelimb mobilization with limb fully retracted.

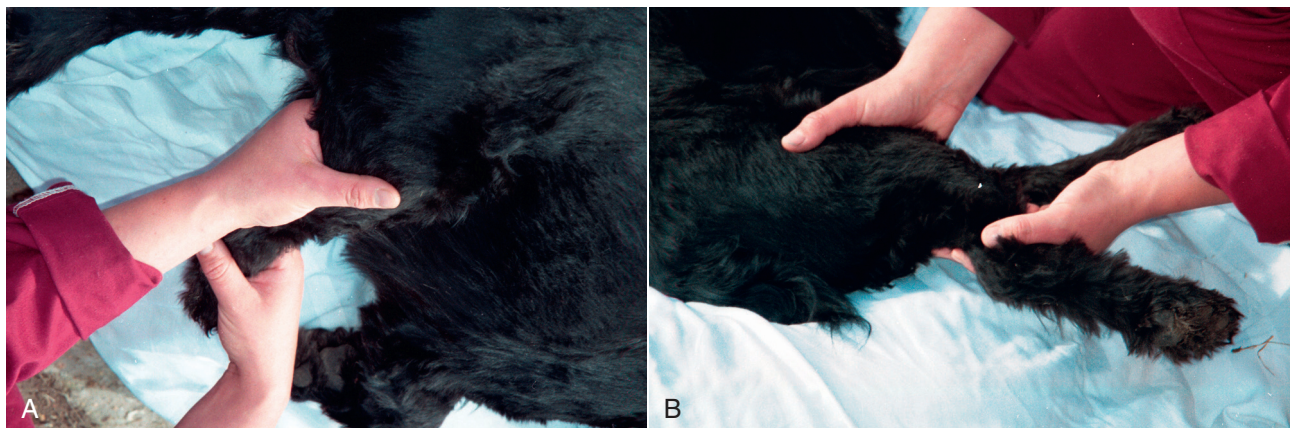


Figure 27-37 A, Hindlimb mobilization with limb fully protracted. B, Hindlimb mobilization with limb fully retracted.

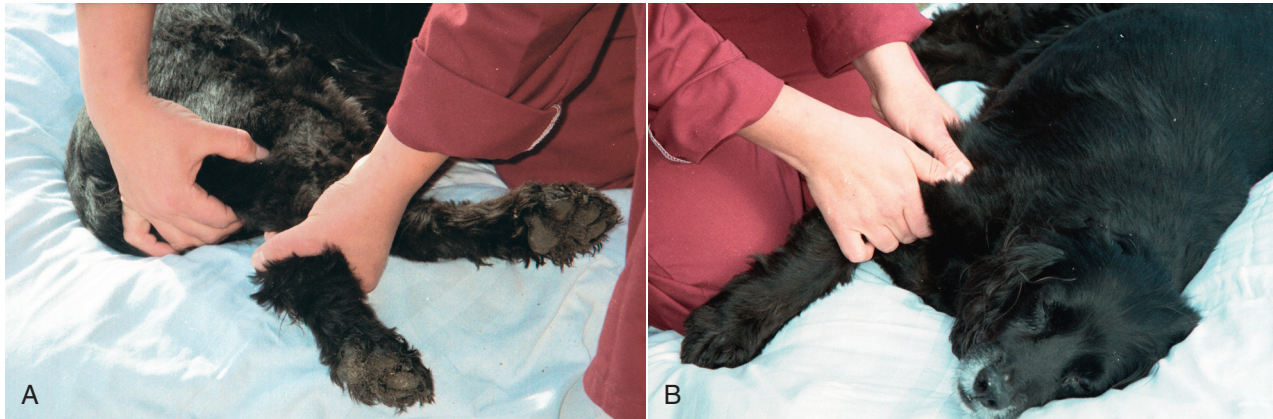


Figure 27-38 A, Isolating muscle spasm while placing rear limb through a range of motion. B, Isolating muscle spasm while placing forelimb through a range of motion.

are performed. Again it is best to work in a sequence, such as from forelimbs to hindlimbs.

Forelimb and Scapular Mobilizations

Place one hand over the scapula and support the forelimb with the other. The scapula is mobilized from elevation to protraction, depression, and retraction. The limb is fully protracted and retracted (Figure 27-36).

Hindlimb

Similar movements are performed on the hindlimb. With one hand supporting the hindlimb and the other resting over the hip, the leg is elevated and depressed before it is protracted and retracted (Figure 27-37).

In some cases when passive movements are performed, areas of spasm may remain within the muscles, restricting full available range of movement. In these situations it is often useful to combine passive movements with trigger point therapy. The isolated muscle spasm is held by the therapist's finger and thumb while the other hand guides the limb through a ROM (Figure 27-38). If restriction occurs again, the position should be held for a minimum of 15 seconds, then released before repeating. Following treatment, the dog should be very relaxed. The dog should be turned over and the massage should be repeated on the other side.

Alternative Methods

Several methods may be used as an alternative or adjunct to massage, including mechanical massagers, grooming the dog with a massager, and using commercially available massage lubricant.

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28 Exercises for Proprioception and Balance

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Many conditions affect proprioception of small animals. The most obvious of these are neurologic conditions. However, recent evidence suggests that musculoskeletal conditions and injuries also affect proprioception, especially of joints.

Exercise enhances balance and proprioception,¹ and a well-designed balance training program can improve balance in normal individuals.²

Balance is the ability to adjust equilibrium at a stance, or during locomotion to adjust to a change in direction or ground surfaces, whereas proprioception is the perception of stimuli produced by the body in relation to movement and changes in body position. Proprioceptive training includes activities that may be performed at low or high speed and require an awareness of limb position in space. Example of activities include walking in circles or a figure eight and walking across obstacles of various shape, height, and spacing. Balance exercises are exercises requiring rapid responses to changes in a surface, such as walking on a trampoline, or standing on a balance or wobble board. Other balance and agility exercises include cavaletti rails, exercise balls, rapid changes of direction while trotting and galloping, ball playing, tug of war, dancing, and wheelbarrowing. This chapter discusses current knowledge regarding proprioception and balance and exercises to enhance patient function.

The Science of Proprioception

The proprioceptive system is responsible for detecting changes in the position of the trunk, limbs, and head. The system is distributed widely throughout all of the spinal nerves, and general proprioception receptors are located in the muscles, tendons, and joints deep to the surface of the body. The receptors respond to changes in the length and position of the structures where they are located, giving the normal individual a precise feeling of where their body parts are located. The postural control system consists of three functional components, including biomechanical, motor coordination, and sensory organization components.

Joint Proprioception

Ruffini and Pacinian receptors have been identified in canine cruciate ligaments and are innervated by axons penetrating from the peripheral synovium surrounding the ligament.³ The proximal third of the ligament contains a

greater number of receptors.⁴ The cranial cruciate ligament is capable of sending afferent input to the central nervous system, and it is important to the biomechanical and proprioceptive functions of the stifle joint, including characteristics of movements and position-related stretches of these ligaments.⁵ The effects of the gamma-muscle-spindle system in the muscles around the knee are so potent that stretching of the cruciate ligaments at relatively moderate loads may induce major changes in responses of the muscle spindle afferents.⁵ These changes may result in changes in the muscular stiffness around the knee joint, helping to affect functional stability of the knee joint.

Mechanoreceptors have been studied in torn and reconstructed anterior cruciate ligaments (ACL) in animals and humans.⁶ In humans with untreated ACL lesions, normal mechanoreceptors remain in the ligament 3 months after injury. The number then gradually decreases, and by the ninth month, few free nerve endings are present.

The correlation between the number of mechanoreceptors in ACL remnants and joint position sense just before an ACL reconstruction in human patients has been studied.⁷ A positive correlation between the number of mechanoreceptors and accuracy of the joint position sense was found, suggesting that proprioceptive function of the ACL is related to the number of mechanoreceptors. Therefore consideration should be given to preserving cruciate ligament remnants during stabilization surgery if the ligament is relatively normal. However, a study in cats that evaluated joint capsule afferent neuron stimulation before and after transection of the cranial cruciate ligament suggested that capsule afferents continue to behave normally after ligament transection, despite the increased joint laxity.⁸

Proprioceptive ability declines with aging in humans.⁹ A randomized clinical trial evaluated the effects of an 8-week balance program on proprioception compared with only a fall-prevention program in healthy older adults (mean age = 73.9).¹⁰ Each group was assessed before and after the study, and a control group also underwent follow-up testing 8 weeks after intervention. Additionally, this study compared the intervention group to a younger, healthy population who were only assessed once for comparison. Interventions included typical balance and proprioceptive exercises such as activities on foam, foam balance beams, wobble board activities, and others. Subjects underwent 1-hour sessions three times per week, for

a total of 8 weeks. The balance group showed significant improvement within the group in velocity discrimination from before and after intervention. In addition, there were significant differences after intervention between both treatment groups. In fact, the balance group showed no significant differences in velocity values after training when compared with younger adults. However, no other differences in joint position sense or threshold to perception of movement were encountered. Because velocity, regulated mainly by muscle spindles, is considered vital to postural stability, this study suggests that proprioceptive training could decrease the risk for falls in older adult humans. However, these results were not maintained long term, suggesting that continued training is necessary.¹⁰

Proprioception is further impaired in older adult patients with knee osteoarthritis (OA), which may contribute to additional functional impairment.^{1,9,11-13} On the other hand, the pain of osteoarthritis does not seem to have good association with disability and joint proprioception.¹⁴ Although patients with arthritis have decreased joint position awareness, physical therapy may improve joint position sense and proprioception through participation in a carefully constructed sensorimotor rehabilitation program in which pain, effusion, and fatigue are minimized.

In healthy individuals, muscle receptors are the primary determinants of joint position sense, and capsular receptors have a secondary role. Studies have been performed in humans to test position sense of the knee joint before and after fatigue.^{15,16} With fatigue, there appears to be a change in mechanism of appreciation of joint position, possibly because of an increased role of capsular receptors regarding joint position sensitivity with muscle fatigue-induced laxity.

Appreciation of joint position was also found to be significantly more sensitive after warming up by jogging and stretching for 4 minutes, demonstrating adequate warm-up prior to activity may be beneficial.¹⁷ Conversely, to test whether cryotherapy affects position sense of the knee, a cooling pad was applied to knees of humans.¹⁸ Cooling for 15 minutes resulted in increased knee joint stiffness and decreased sensitivity of joint position sense. However, another study indicated that cryotherapy of the shoulder does not affect joint position awareness.¹⁹ It may be possible that differences exist in different joints, with more superficially located joints affected more by cryotherapy. These findings should be considered in therapeutic programs that involve exercise immediately after applying cryotherapy.

Long-term effusions and the nature of the inflammatory fluid may be responsible for the seeming loss of proprioception observed in some clinical conditions with chronic joint effusion. However, infusion of saline to simulate acute joint effusion apparently does not change proprioceptive ability.²⁰ An elastic bandage placed on the knee of humans had a positive effect on joint proprioception, and may be recommended in those patients with joint proprioception problems.^{21,22} Patella taping techniques improved

joint position sense in patients with patellofemoral pain syndrome that had poor proprioception to begin with, but did not help those with good proprioception.²³

Neurologic Conditions

Proprioception is commonly affected by neurologic conditions in animals and people. Proprioception and rehabilitation have been studied in people following a stroke. There are many approaches to physical therapy following stroke injury in people. One method of treatment, known as supported treadmill training, appears to have beneficial effects on proprioceptive function in people after a stroke.^{24,25} A 4-week period of body weight–supported treadmill training with up to 40% of the patient's body weight supported resulted in significantly improved clinical outcomes, including functional balance, motor recovery, overground walking speed, and overground walking endurance.²⁶ The effects lasted for at least 3 months after training. The subjects with greater gait impairments benefited the most from training with body-weight support.

Another study evaluated the delayed effects of a balance training program compared with no training on patients following a hemiplegic stroke.²⁷ Dynamic balance function of patients in the visual feedback training group had significant improvements when compared with the control group. Activities of daily living also had significant improvements at 6 months of follow up in the trained group. A study of older humans indicated that a short individualized exercise program improved functional balance. This improvement was maintained for at least 1 month, but there was no effect 1 year after training ceased, suggesting that continued training may be necessary.

Strength training should also be considered as part of the rehabilitation program for patients. In one study, chronic human stroke patients underwent a progressive resistance strength training program to evaluate changes in muscle strength, gait, and balance.²⁸ Subjects participated in a 12-week resistance training program. Lower limb strength improved 68% on the affected side. Repeated chair stand time improved 21%, motor performance improved 9%, and static and dynamic balance improved 12%. Other studies have confirmed the benefits of strength training in human stroke patients.²⁹ These results suggest that strength training should be considered even in patients with chronic deficits.

Proprioception and Activity

In the past decade, research in humans has suggested that stretching prior to activity may reduce acute proprioception, which might lead to injury. A recent study showed significant differences in detecting knee movement after undergoing a hold-relax proprioceptive neuromuscular facilitation stretch routine on the hamstrings and quadriceps compared with a control group.³⁰ The treated group had a significantly slower reaction time from pretest to posttest, whereas the control group did not. It is important

to note that this study suggests stretching in general may promote decreased proprioception in certain joints immediately after stretching. This could hold implications for dogs in the late phases of rehabilitation, especially for sporting dogs, in which stretching prior to training could alter coordination and motor function, considering that proprioception is essential for skilled movements.

Proprioceptive training may help reduce injuries in athletes. In a recent randomized controlled trial of highly competitive basketball players, a treated group ($n = 81$), who received a proprioceptive training regimen along with normal workouts, was compared with athletes undergoing a normal training routine ($n = 91$).³¹ Proprioceptive training was performed once a week and consisted of six stations, each performed twice each for 45 seconds, followed by 30 seconds of rest as they switched stations. All exercises were clearly illustrated and showed proper progression. Over the course of an entire season, 21 injuries occurred in the control group compared with 7 in the treated group. The treated group also showed a more stable single limb stance in the frontal plane and overall sway compared with the control group. Although this study was performed on humans, it suggests that proprioceptive training may prevent injuries, and this might also be applicable to animal rehabilitation.

Finally, in a recent review to help guide clinical decisions on prevention of ACL injury with the use of proprioceptive and balance training, the authors concluded that there is a lack of solid evidence to determine if this type of training has beneficial effects. However, this was mainly a result of poor experimental design of the studies (such as poor description of training regimens and high drop-out rates), which at least determined no detrimental effects of proprioceptive training.³² Others have agreed, stating that many articles combine proprioceptive exercises with plyometrics, agility, and sport-specific exercises, making the specific effects of each individual treatment difficult to ascertain.³³

Proprioceptive and Balance Training

When an animal is able to safely stand (independently or with assistance), activities to improve balance may begin. Balance may be assessed using a scoring system based on similar evaluation instruments in people (Table 28-1). Static balance is the animal's ability to maintain balance while standing, while dynamic balance is the ability to maintain balance while the body is moving. Static balance training is generally performed with the animal in a standing position, either independently or with support. The following exercises may be performed to challenge the animal's balance. These exercises should be conducted on a nonslip surface to reduce the risk of falling.

Body Slings

Patients with weakness, limited motor function or poor proprioception may benefit from placement in a body sling

Table 28-1 Balance Score for Small Animals

BALANCE ITEM SCORE (0-5)

- | | |
|---|-------|
| 1. Sitting squarely and unsupported | _____ |
| 2. Change of position: sitting to standing | _____ |
| 3. Change of position: standing to sitting | _____ |
| 4. Position from lateral recumbency to sternal position | _____ |
| 5. Standing unsupported | _____ |
| 6. Standing with feet together | _____ |
| 7. Standing with one leg raised | _____ |
| 8. Turning 360 degrees | _____ |
| 9. Stair climbing | _____ |
| 10. Balancing while standing and turning head | _____ |

TOTAL (0-50):

Score each item from 0 to 5,

0 – unable to perform the activity

1 – barely performs the activity with assistance

2 – performs activity to some extent with difficulty, with or without assistance

3 – performs activity reasonably well with mild difficulty and no assistance

4 – performs activity well with minimal difficulty and no assistance

5 – performs the activity normally

INTERPRETATION

0-15: Generally needs full support

16-40: Generally can walk with assistance

41-50: Generally independent activity

Based on the following in people:

Berg K, Wood-Dauphinee S, Williams JL, Maki B: Measuring balance in the elderly: Validation of an instrument. *Can J Pub Health* July/August supplement 2:S7-S11, 1992.

Berg K, Wood-Dauphinee S, Williams JL, Gayton D: Measuring balance in the elderly: preliminary development of an instrument. *Physiother Can* 41:304-311, 1989.

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or harness to provide support for standing (Figure 28-1). An advantage of a sling is that the patient is able to maintain a position with the limbs placed under the body in a standing position, which may provide an opportunity for limb strengthening and early proprioceptive training. Frequent episodes of supported standing also help relieve pressure on bony prominences and may reduce the chances of decubital ulcer formation. Excessive congestion of the lungs that may result from spending long periods in lateral recumbency may be prevented with early implementation of standing in a sling. A body harness might be used for patients with tetraparesis, but must be adjusted properly so that respiration is not compromised and the limbs are not compressed in the openings of the harness. Proper padding is also important



Figure 28-1 A, Patients with poor balance or proprioception may benefit from placement in a body sling or harness to provide support for standing. B, An elastic band may also be used for support and to provide a form of rhythmic stabilization while standing or walking.



Figure 28-2 Rubber bungee cords may be used to secure a harness to the supporting frame.



Figure 28-3 For weight shifting of the rear limbs, the therapist is positioned behind the patient, and the hands are placed on either side of the pelvis for support.

for comfort. The skin should be inspected after each session in the sling or harness to identify areas of potential skin irritation or breakdown. Rubber bungee cords may be used to secure a harness to the supporting frame (Figure 28-2). This may aid in the facilitation of weight bearing and movement during the sessions. The elasticity of the cord permits some gentle up-and-down movement that encourages the animal to safely bear partial weight on its limbs. It also may allow some attempts to move forward, backward, and side to side. A gentle bouncing motion may be manually introduced by the therapist to initiate rhythmic stabilization of muscle groups throughout the trunk and limbs.

Weight Shifting

The therapist may begin gentle weight-shifting activities by supporting the animal on either side, and then gently pushing on one side, then pushing back the other way in a slow, rhythmic fashion. For example, for weight shifting of the rear limbs, the therapist is positioned behind the patient, and the hands are placed on either side of the pelvis (Figure 28-3). The animal is slowly pushed to one side, while supporting the patient with the other hand. The animal is then slowly pushed back the other way. As balance improves, the speed of the unbalancing movements may increase. Additional challenges may be added



Figure 28-4 A supporting towel is moved back and forth to shift the dog's weight ("shoe buff" maneuver).



Figure 28-5 Perturbation exercises are performed to disturb the animal's balance by gently pushing the animal at the hips or shoulder without supporting the opposite side.

by slowly moving a supporting towel or sling back and forth to shift the dog's weight ("shoe buff" maneuver) (Figure 28-4).

As balance improves, the handler may perform *perturbation exercises* to disturb the animal's balance by gently pushing the animal at the hips or shoulder without supporting the opposite side (Figure 28-5). The goal is to disturb its balance just enough so the animal can recover, being careful not to push with a force that may cause the animal to fall. Generally, pushing the animal to the more affected



Figure 28-6 A loaded weight shift is performed by placing force over the dorsal aspect of the dog's body while shifting the weight to the affected side.

side challenges the animal sufficiently to allow the activity to have the desired effect.

In situations of reduced weight bearing on one side, a *loaded weight shift* may be performed (Figure 28-6). Dogs recovering from joint surgery or other conditions that have reduced weight bearing may benefit from this exercise to encourage some weight bearing, early muscle strengthening, and joint position awareness. For this exercise the therapist's forearm that is on the side of the injury is placed across the animal's dorsal pelvis (hindlimbs) or shoulders (forelimbs) and gentle downward pressure is applied. With the other hand, the weight is gently shifted toward the affected side.

Some dogs become conditioned to having the therapist push on one side to shift the weight to the other side, and shift their weight back toward the therapist to prevent being pushed toward the affected side. In this case, a *rebound weight shift* may be effective. For this maneuver, the therapist gently pushes the animal toward the affected side as usual. When the animal shifts its weight to resist the movement, the therapist suddenly releases pressure, and simultaneously pushes gently toward the unaffected side (Figure 28-7). This results in a sudden unbalancing; the animal initially shifts its weight toward the unaffected side, but to keep from falling, it immediately shifts its weight back toward the affected side.

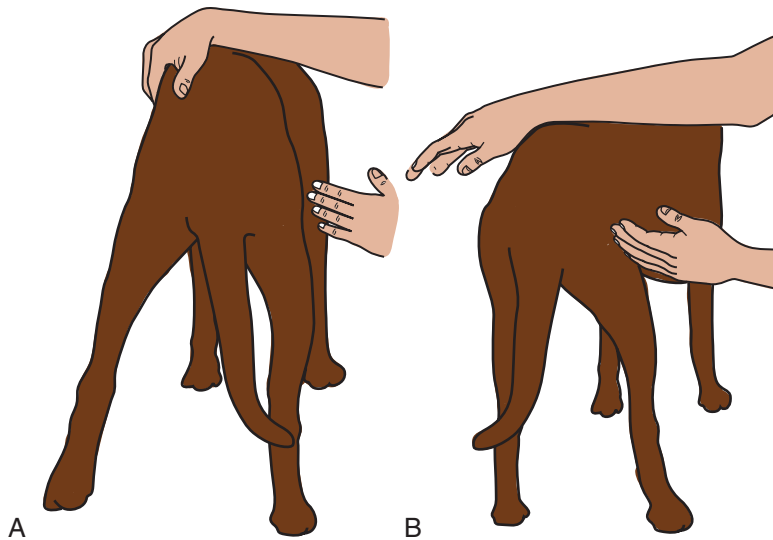


Figure 28-7 Rebound weight shift to encourage an animal to bear weight on the affected limb. The dog's weight is initially shifted toward the normal (*right*) limb. Pressure is suddenly released, resulting in a shift in weight toward the affected (*left*) limb to maintain balance.



Figure 28-8 The therapist encourages weight shifting to help develop improved proprioception by moving a treat side to side.

While the animal is standing, a treat or ball may be used to encourage weight shifting. The dog will follow the treat up and down and side to side (Figure 28-8). Start with small movements and progress to larger, more challenging movements. The movement of the head causes the dog's center of gravity (COG) to shift. As the COG shifts, the dog must shift its weight to maintain its balance. To maintain the unassisted standing position, the animal is required to use strength, coordination, and balance. If the dog is motivated by ball play, a more challenging form of this exercise is close-distance ball tossing from above and the sides. During early attempts at these exercises it may be necessary to have someone available for standby assistance as the dog may be overchallenged, lose its balance, and fall



Figure 28-9 The therapist is ready to provide support to a patient in the event of a loss of balance.

(Figure 28-9). As the patient improves, close-distance ball tossing may be performed with the patient on a trampoline or air mattress.

Manual unloading of a limb during stance is another challenge to balance. Lifting and holding a single limb off the ground while the dog is standing cause a shift in the animal's COG (Figure 28-10). The animal shifts its body weight and COG to maintain the standing position. During a session, the handler may lift each leg separately to see where the animal is the weakest and focus on that area in successive sessions. If the animal is unable or unwilling to perform this exercise, it will not shift its weight properly,